Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by relative deadwood class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (104 sites, 381 site-years) · deadtrees v4.1

ΔRMSE = RMSE(with M) − RMSE(without M)

XGBoost (Optuna) · 12m — 66% improved
XGBoost (Optuna) · 24m — 56% improved
LightGBM (Optuna) · 12m — 63% improved
LightGBM (Optuna) · 24m — 58% improved

GPPsat (µmol m⁻² s⁻¹)

NEPmax (µmol m⁻² s⁻¹)
76% improved | 70% improved | 73% improved | 74% improved

ETmax (mm d⁻¹)
56% improved | 61% improved | 54% improved | 54% improved

WUE (g C mm⁻¹)
53% improved | 55% improved | 54% improved | 56% improved

Sites (sorted by ΔRMSE within each panel)

Relative deadwood
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 9.4 %, 41 sites)
- Mid (9.4–20.6 %, 40 sites)
- High (> 20.6 %, 23 sites)